STAT/MA 41600
In-Class Problem Set #32: October 30, 2026
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Problem Set 32 Answers

1. The probability is $\int_0^\infty \int_x^\infty \frac{1}{40} e^{-x/40} \frac{1}{45} e^{-y/45} \, dy \, dx = 9/17$.

2. We compute:

$$\begin{aligned} P(X_3 > X_2 > X_1) + P(X_3 > X_1 > X_2) &= \int_0^\infty \int_{x_1}^\infty \int_{x_2}^\infty \frac{1}{2} e^{-x_1/2} \frac{1}{2.5} e^{-x_2/2.5} \frac{1}{3} e^{-x_3/3} \, dx_3 \, dx_2 \, dx_1 \\ &\quad + \int_0^\infty \int_{x_2}^\infty \int_{x_1}^\infty \frac{1}{2} e^{-x_1/2} \frac{1}{2.5} e^{-x_2/2.5} \frac{1}{3} e^{-x_3/3} \, dx_3 \, dx_1 \, dx_2 \\ &= 90/407 + 36/185 = 846/2035 = 0.4157 \end{aligned}$$

3. We calculate $\int_{2.25}^\infty \int_{2.25}^\infty \int_{2.25}^\infty \frac{1}{2} e^{-x_1/2} \frac{1}{2.5} e^{-x_2/2.5} \frac{1}{3} e^{-x_3/3} \, dx_3 \, dx_2 \, dx_1 = e^{-111/40} = 0.06235$.

4. The probability is $\int_0^2 \int_0^u (\frac{1}{2})(\frac{1}{1.4}) e^{-x/1.4} \, dx \, du = \frac{3}{10} + \frac{7}{10} e^{-10/7} = 0.4678$.

5. For any individual person who drinks the witches' brew, the probability that they begin to feel the effects within the first 2.5 minutes is $\int_0^{2.5} \frac{1}{2} e^{-x/2} \, dx = 1 - e^{-5/4} = 0.7135$.

The number of people who begin to feel the effects within the first 2.5 minutes is a Binomial random variable with $n = 4$ and $p = 1 - e^{-5/4} = 0.7135$.

So the probability that at least 3 out of the 4 people begin to feel the effects of the witches' brew within the first 2.5 minutes is

$$\binom{4}{3} p^3 (1-p)^1 + \binom{4}{4} p^4 (1-p)^0 = 4(1 - e^{-5/4})^3 (e^{-5/4})^1 + (1 - e^{-5/4})^4 (e^{-5/4})^0 = 0.6754.$$